ARC Research · Weekly AI update · 20–26 Oct 2026

Cooling the AI GPU

Single-phase vs two-phase liquid cooling

Data as of 26 Oct 2026 · Educational purposes only, not investment advice

Agenda

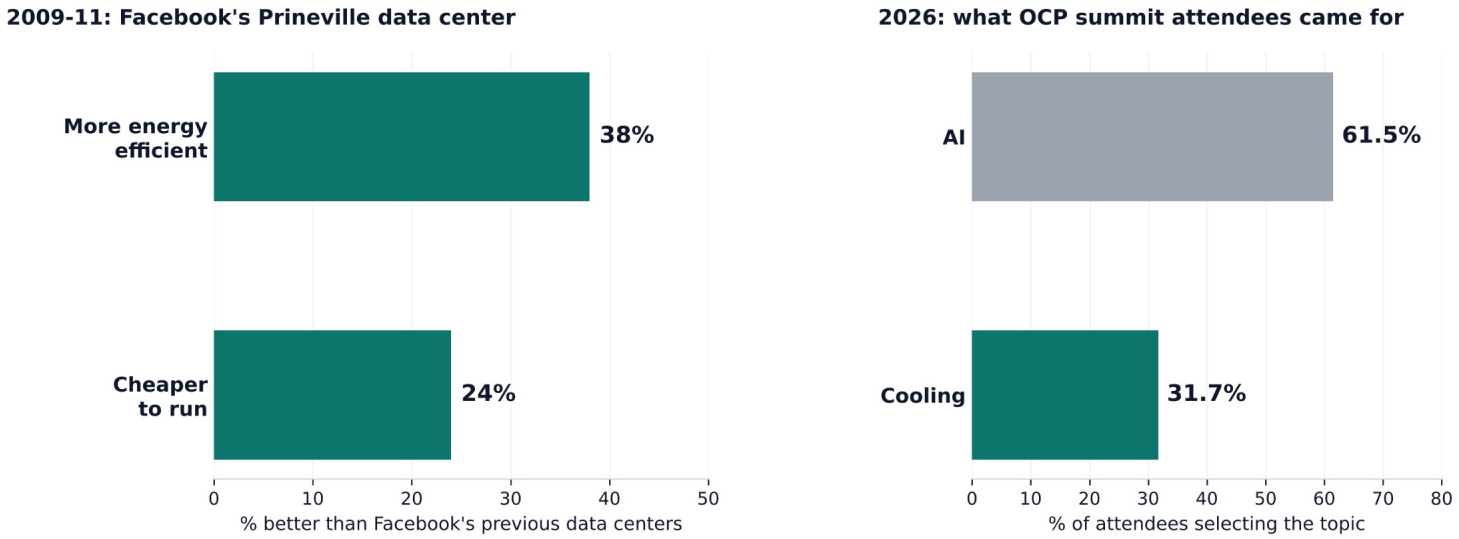
01	**The heat problem**	GPU and rack power keep climbing
02	**OCP’s cooling playbook**	Workstreams, specs, who says what
03	**Single-phase cooling**	The coolant stays liquid
04	**Two-phase cooling**	The coolant boils at the chip
05	**Head-to-head**	Comparison and the ARC view
06	**The market angle**	Who pays, who profits, who’s buying

What is the Open Compute Project?

ARC RESEARCH | AI UPDATE | DATA CENTER COOLING

OCP began as one efficient data center, now it shapes AI cooling

Facebook's Prineville build was cheaper to run; cooling is now the No. 2 topic at OCP's summit after AI



Source: opencompute.org/about (Prineville vs Facebook's previous facilities, as stated by OCP; company-reported, not independently audited);
OCP, '2026 OCP Global Summit by the Numbers' (20 Oct 2026): share of attendees selecting each interest; only the two top interests are shown.

Open-source hardware for data centers

Started by Facebook (now Meta) in 2011

Members share specs anyone can build

What happened

THE THING

OCP is standardizing liquid cooling for AI

NVIDIA put its fully liquid-cooled Vera Rubin rack into OCP this month

THE APPETITE

Racks heading from 120 kW to 600 kW+

Air cooling strains past ~20 kW a rack

THE CATCH

Two-phase cools harder, but its fluids are at risk

PFAS phase-outs hit the classic two-phase fluids

01

The heat problem

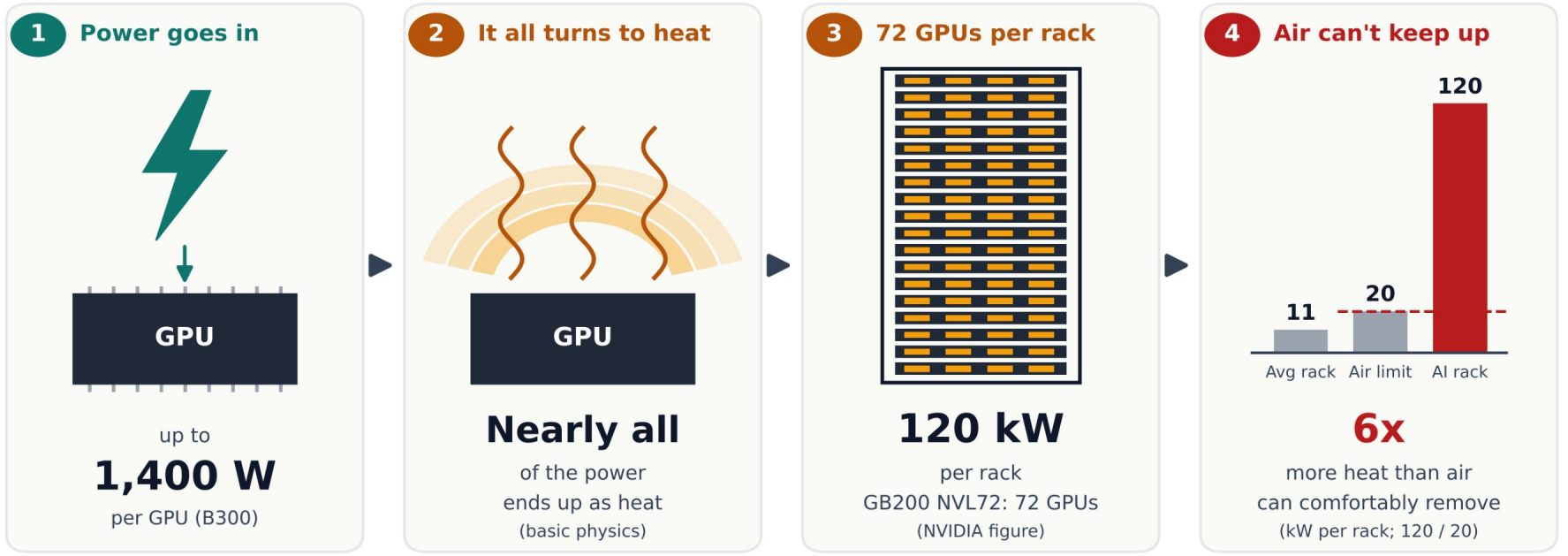
Up next: How heat arrives · GPU power · Rack power · What heat costs

How the heat problem arrives

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How the heat problem arrives, in four steps

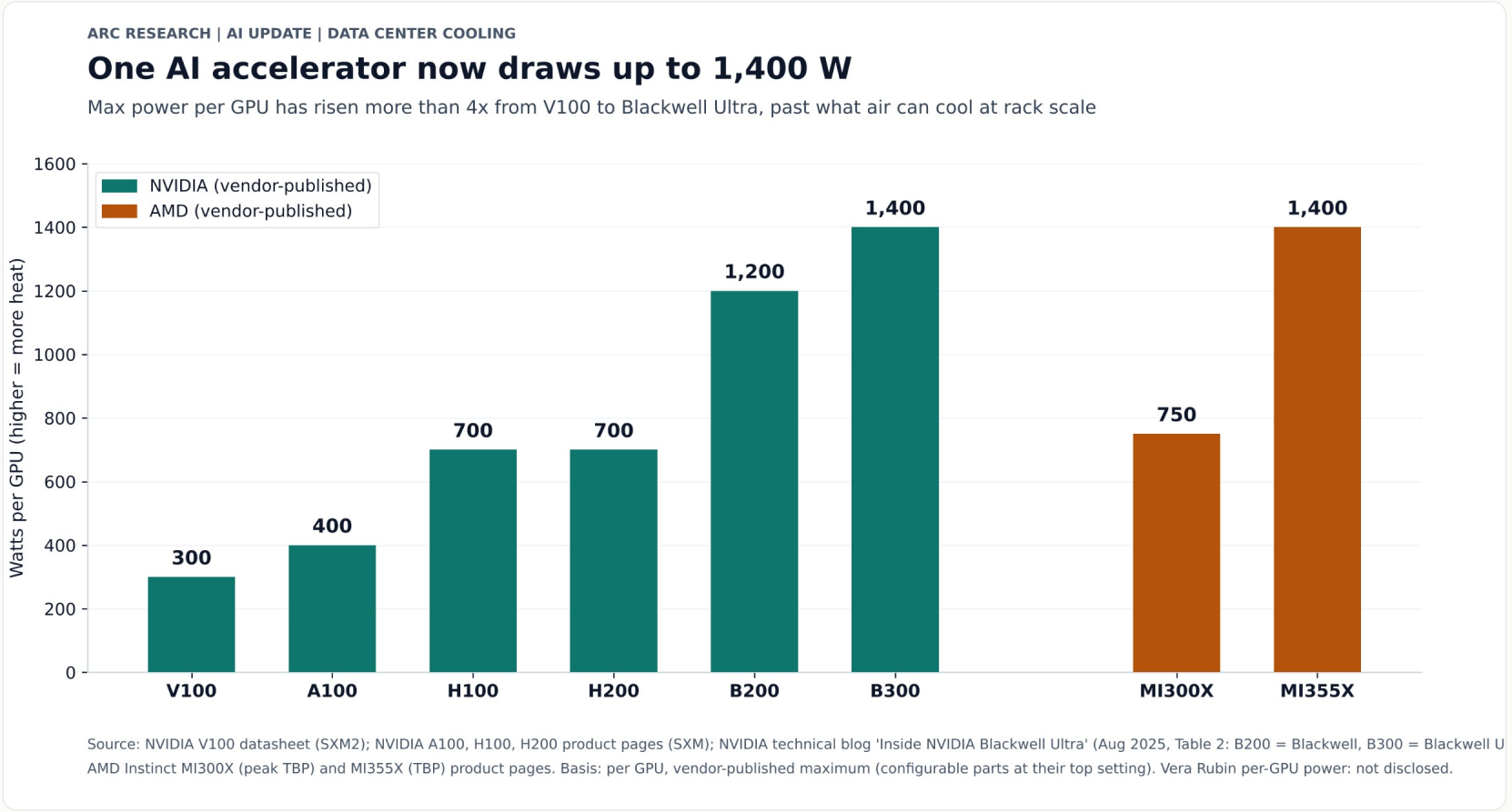
Power in becomes heat out; packing 72 GPUs into one rack makes more heat than air can carry away



If the heat stays: the chip hits its slowdown temperature and throttles, and in training one slow GPU holds back the cluster

Sources: NVIDIA 'Inside Blackwell Ultra' blog, Aug 2025 (B300 up to 1,400 W per GPU); NVIDIA blog, Apr 2025 (GB200 NVL72: 72 GPUs, 120 kW per rack); Uptime Institute Global Survey 2026 (average rack ~11 kW); Uptime Cooling Systems Survey 2024 (air too costly above ~20 kW, operator view); NVIDIA NVSentinel runbook (thermal slowdown). '6x' = ARC Research calculation, 120 / 20. Schematic, not to scale.

One GPU now draws up to 1,400 W



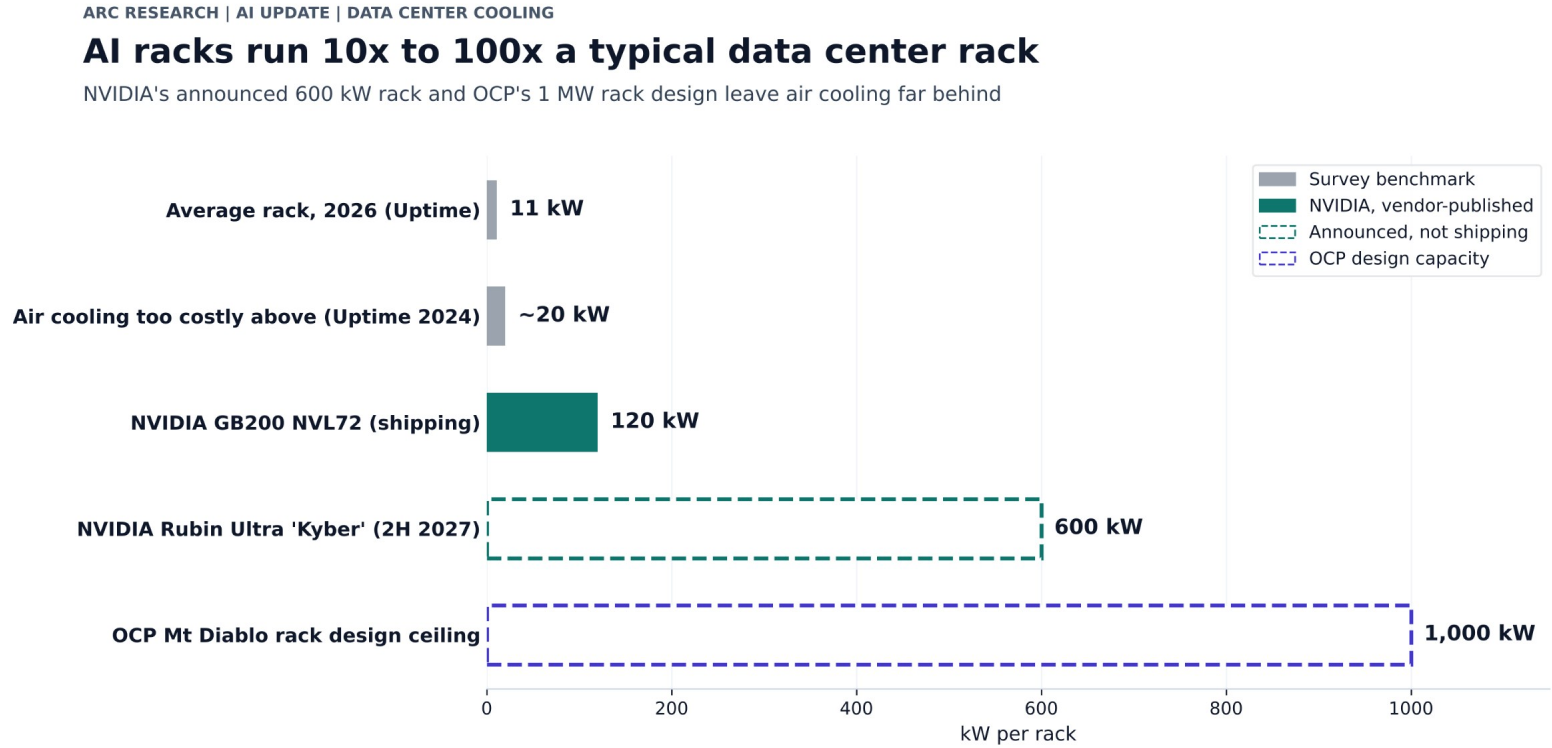
AMD

More than 4× per GPU since V100

NVIDIA and AMD both top out at 1,400 W

Nearly all of it ends up as heat

AI racks run 10× to 100× a normal rack



Today’s AI rack: 120 kW

NVIDIA’s 2027 rack: 600 kW

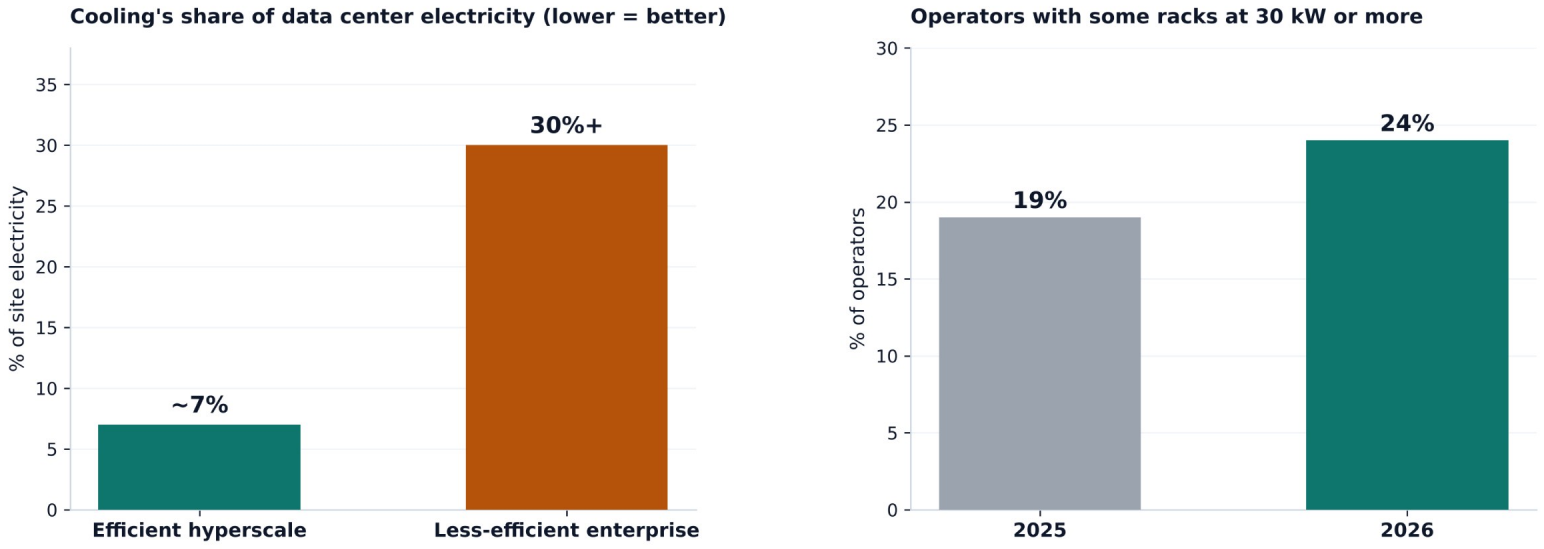
OCP is designing for 1 MW

Heat costs speed and electricity

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Cooling eats power, and dense racks are spreading

Inefficient sites spend over 30% of electricity on cooling; a quarter of operators already run 30 kW+ racks



Source: IEA, Energy and AI (2025): cooling about 7% of electricity in efficient hyperscale sites to over 30% in less-efficient enterprise sites (30% plotted as a lower bound).
Uptime Institute Global Data Center Survey 2026 (fielded Apr-May 2026): share of respondents with some racks at 30 kW or above, 2025 vs 2026.

Too hot, and a GPU slows itself down

One slow GPU stalls a whole training step

Poor cooling wastes 30%+ of power

02

OCP’s cooling playbook

Up next: Five workstreams · The phase × contact map · Who says so

OCP Cooling Environments: five workstreams

Cold plate

Direct-to-chip loops, fluids, connectors

CDU

Coolant distribution unit designs

Immersion

Tanks, fluids, server design

Door HX

Rear-door heat exchangers

Heat reuse

Designs, economics, policy

2018

Advanced Cooling Solutions launched

2019

Immersion requirements Rev 1.0

2025

Mt Diablo rack · Deschutes CDU spec

2026

Vera Rubin rack submitted · cold plate test spec

Two questions place every approach

	Single-phase · stays liquid	Two-phase · boils
Cold plate Fluid in a plate on the chip	**Water or PG25 loop** OCP coolant specs · loop Rev 2 · Deschutes CDU **Mainstream: GB200, GB300, Vera Rubin**	**Dielectric fluid boils in the plate** OCP two-phase cold plate guidelines v1.1 **Emerging**
Immersion Whole server in fluid	**Oil bath, pumped to a heat exchanger** OCP immersion requirements **Niche but proven**	**Sealed tank, vapor condenses on a coil** OCP immersion requirements · two-phase fluid spec v1.0 **Fluid-supply risk**

Who says so

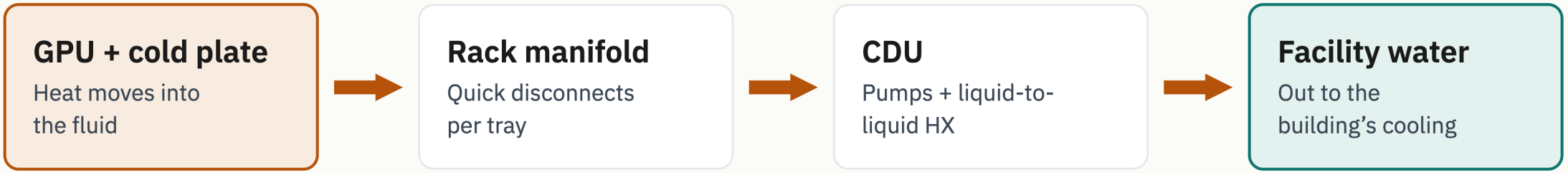
Claim	Source	Status
GB200 NVL72 rack draws 120 kW	NVIDIA blog	Vendor-published
Vera Rubin: 45°C single-phase cooling	NVIDIA Rubin blog	Vendor-published
Rubin Ultra “Kyber” rack: 600 kW	Jensen Huang via DCD	Announced, not shipping
Two-phase PUE beats single-phase	GRC (single-phase vendor)	Vendor claim
2P50 fluid cuts cooling energy 90%+	Chemours	Vendor claim, developmental
Max GPU power two-phase can cool	OCP specs	Not published
EU exemption for two-phase fluids	None found	Not addressed anywhere

03

Single-phase cooling

Up next: How the loop works · Where it fits

Single-phase: the coolant just warms up



Cooled liquid returns to the plate in a closed loop

Heat = flow × cp × ΔT

More heat needs more flow or a bigger temperature rise

45° C

Vera Rubin coolant supply: warm water, single-phase

Single-phase: today’s default for AI racks

Strengths

- Water-glycol: cheap, PFAS-free
- Open OCP specs, many suppliers
- Warm water cuts chiller use
- NVIDIA’s chosen path for Vera Rubin

Limits

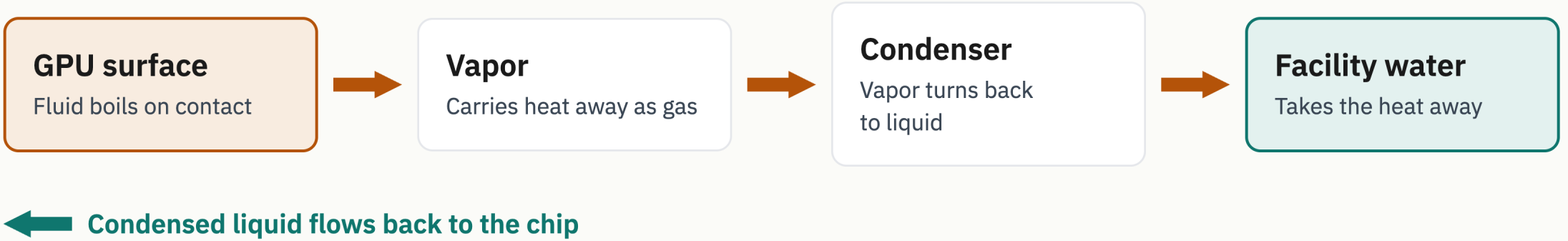
- Flow must rise with every watt
- Water near electronics: leak risk
- Plates cool only what they touch
- Pumps and CDUs keep growing

04

Two-phase cooling

Up next: How boiling cools · The fluid problem

Two-phase: the coolant boils on the chip



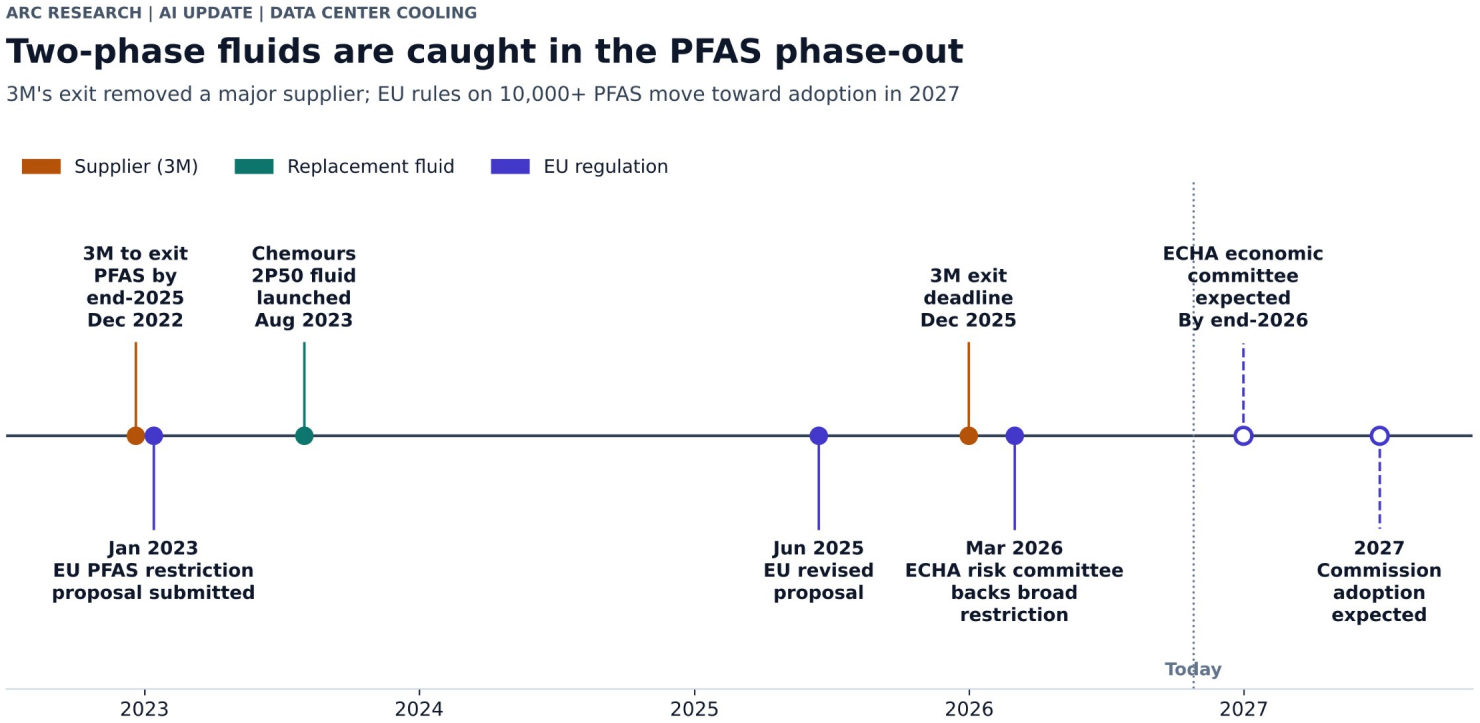
Boiling holds the temperature

Heat goes into changing state, not raising temperature

49° C

Boiling point of Chemours' 2P50 fluid (water: 100°C)

Two-phase fluids meet the PFAS phase-out



3M

3M, a major fluid maker, has exited

EU rules target 10,000+ PFAS

New fluids exist, but are still early

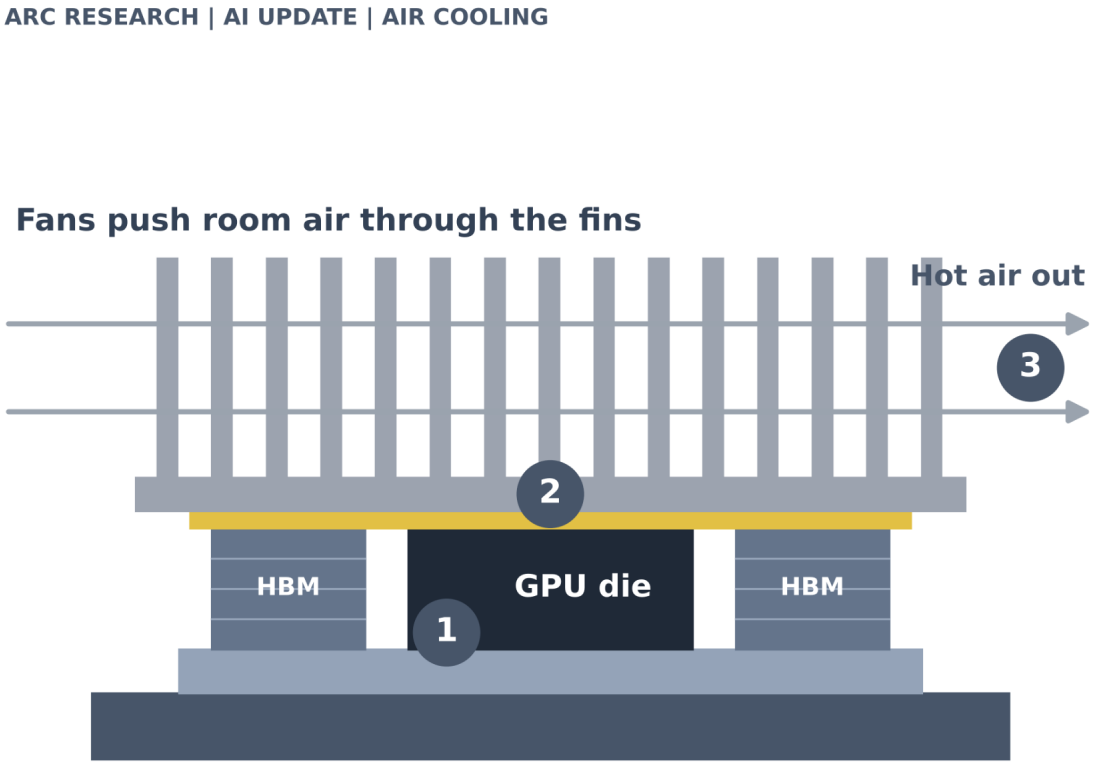
05

Head-to-head

Up next: Four ways to cool a GPU · Side-by-side · Who’s building it · What makes it hard · The ARC view

Way 1 of 4 · Air: heatsink and fans

1 Air | 2 Single-phase plate | 3 Two-phase plate | 4 Immersion



Schematic - not to scale, not a die shot

1. The GPU die turns power into heat
2. Thermal paste moves it into a finned heatsink
3. Fans blow air through the fins and out

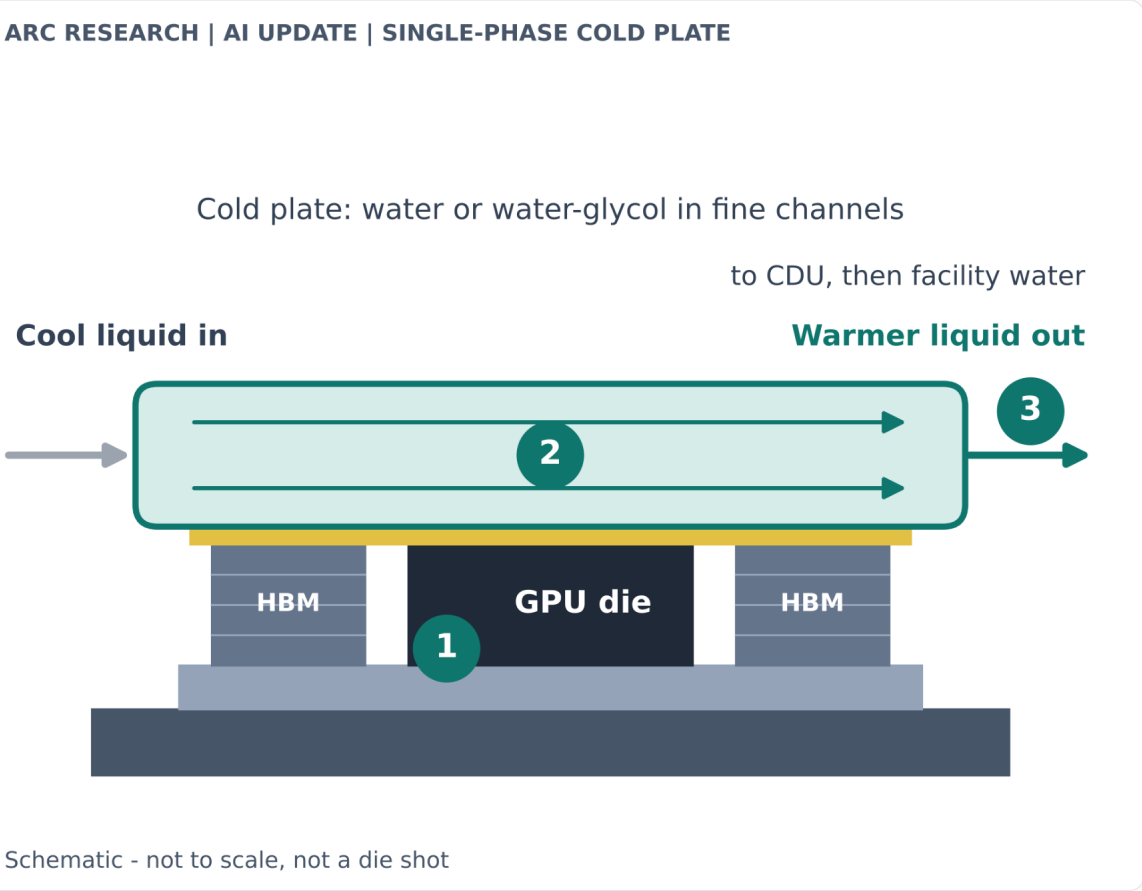
Used in: DGX B200

8 GPUs in one server, ~14.3 kW, air-cooled

Limit: air strains above ~20 kW per rack

Way 2 of 4 · Single-phase cold plate

1 Air | **2 Single-phase plate** | 3 Two-phase plate | 4 Immersion



1. The GPU die turns power into heat
2. Liquid in the plate’s channels absorbs it and warms up
3. Warm liquid goes to the CDU, then building water

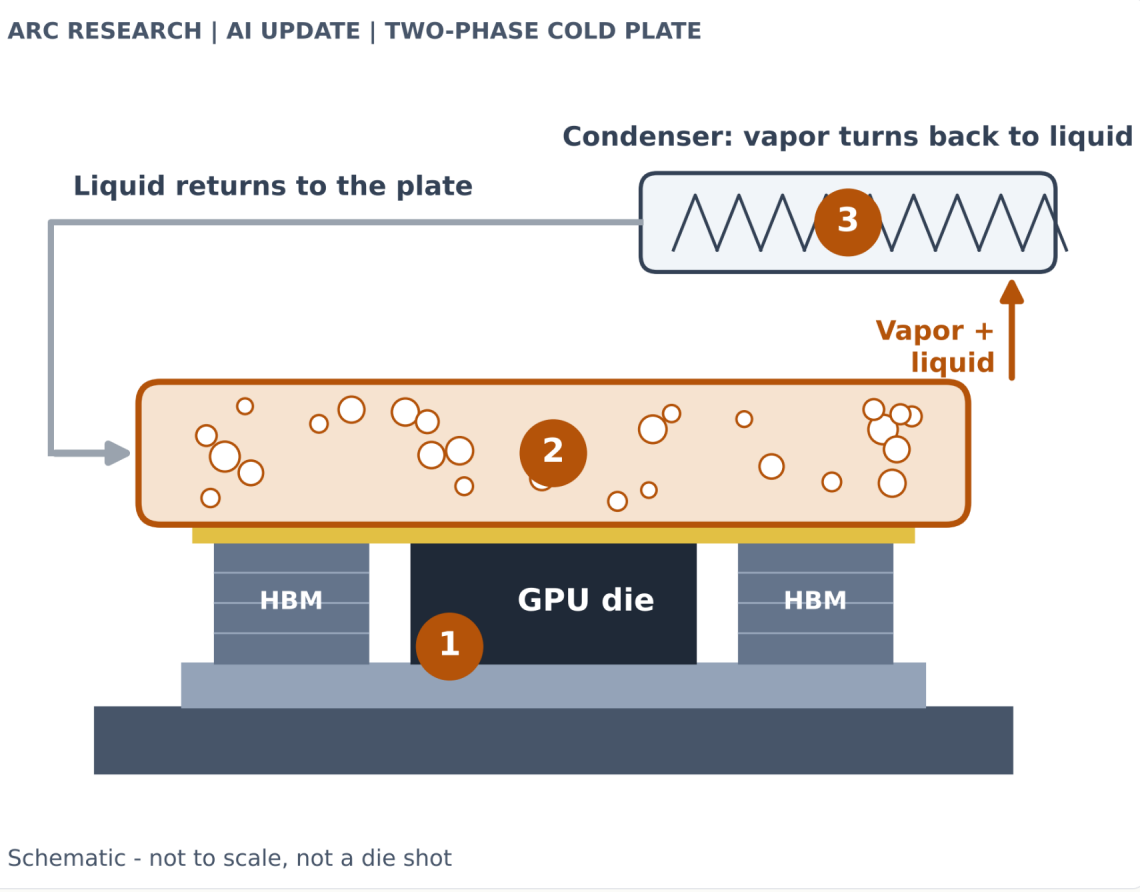
Used in: NVIDIA’s AI racks

GB200 NVL72 (120 kW), GB300 NVL72, Vera Rubin

Vera Rubin runs on 45°C supply water

Way 3 of 4 · Two-phase cold plate

1 Air | 2 Single-phase plate | **3 Two-phase plate** | 4 Immersion



1. The GPU die turns power into heat
2. Fluid in the plate boils, soaking up heat
3. A condenser turns vapor back to liquid

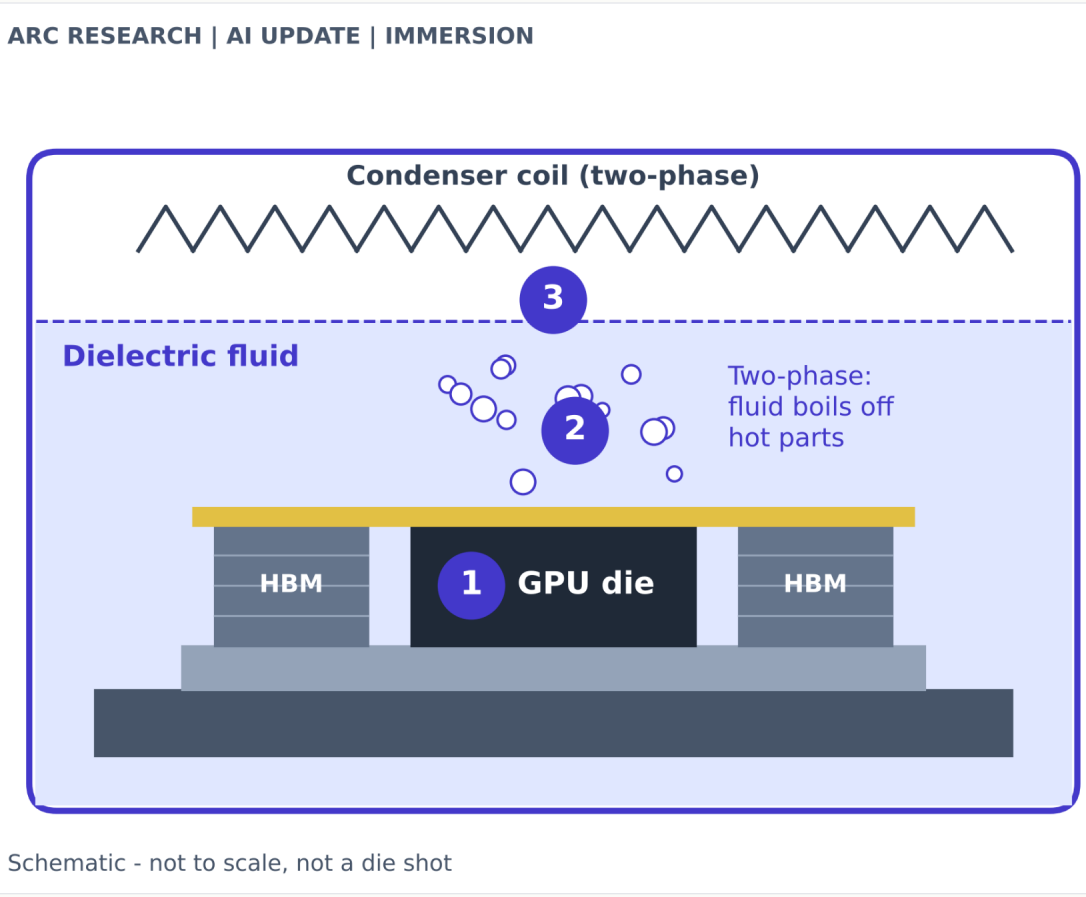
Status: emerging

OCP guidelines v1.1; no NVIDIA design found

Chemours 2P50 fluid boils at 49°C

Way 4 of 4 · Immersion

1 Air | 2 Single-phase plate | 3 Two-phase plate | **4 Immersion**



1. The whole server sits in non-conductive fluid
2. Oil flows past hot parts, or fluid boils off them
3. Heat exits via a heat exchanger or condenser coil

Status: niche but proven

OCP immersion requirements since 2019

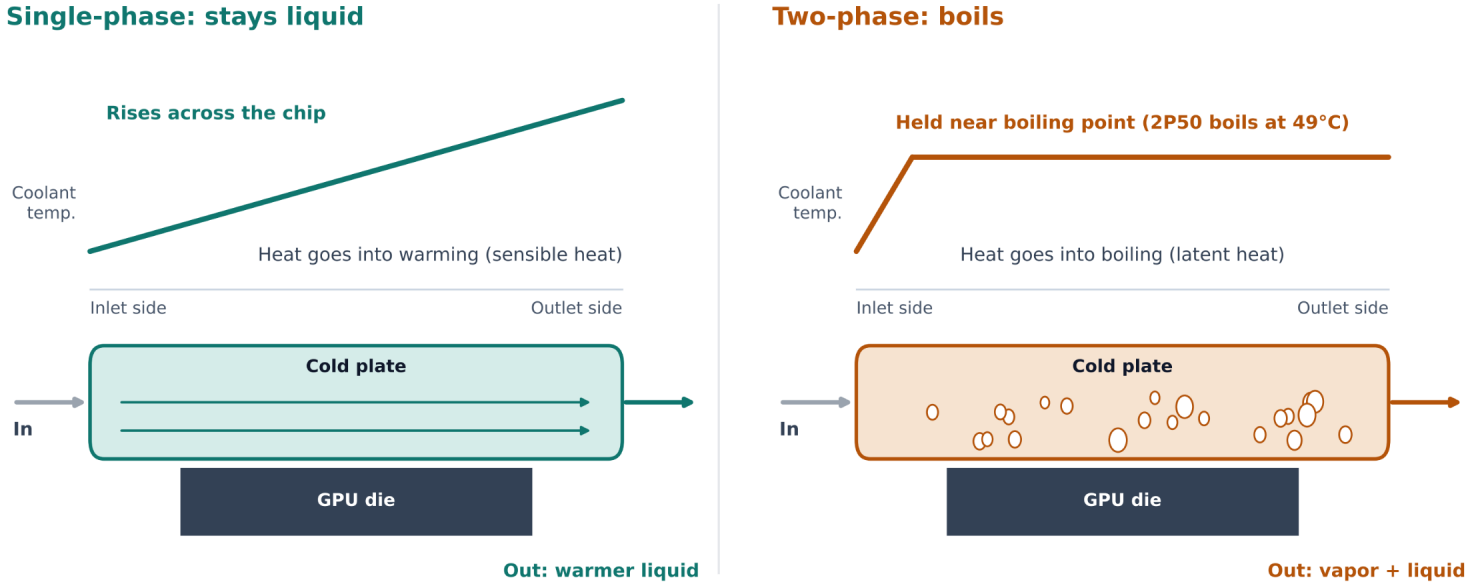
Oils must have a flash point above 150°C

Single-phase vs two-phase, side by side

ARC RESEARCH | AI UPDATE | DATA CENTER COOLING

Same cold plate, two ways to carry the heat

Single-phase coolant keeps warming as it crosses the chip; two-phase fluid boils and stays near its boiling point



Schematic - not to scale, not a measurement. Curves show the shape of coolant temperature along the plate, not measured values.

Sources: OCP ACS Immersion Requirements Rev 1.0 (phase definitions); NVIDIA Rubin platform blog, Jan 2026 (45°C single-phase supply); Chemours Opteon 2P50 page (49°C boiling point).

Single-phase: simple, PFAS-free, NVIDIA's pick

Two-phase: steadier chip temperature

Two-phase risk: fluid supply and PFAS rules

Who’s building it

Chipmakers

NVIDIA
Liquid-cooled racks; Vera Rubin design given to OCP

AMD
1,400 W MI355X

Hyperscalers

Google
Deschutes CDU, open-sourced

Meta, Microsoft
Mt Diablo 1 MW rack

Microsoft
Cooling inside the chip (prototype)

Single-phase suppliers

Vertiv · Schneider
Racks, CDUs, chillers

Eaton · Ecolab
Bought Boyd, CoolIT

Cooler Master · Nidec
MW-scale CDUs

Two-phase bets

Accelsius
Claims 4,500 W chips in tests

ZutaCore
Two-phase cold plates

Chemours
2P50 fluid, still developmental

Immersion

GRC
Single-phase tanks

Submer
Building a 56 MW site

LiquidStack
Bought by Trane

What makes it hard

Buildings built for air

Average rack is ~11 kW; liquid needs new pipes, CDUs, water

Liquid near electronics

Leaks, connectors, shipping racks full of coolant

Standards and cost

Parts must interconnect; OCP specs are still arriving

Scaling the supply chain

CDUs now sized in megawatts; buyers racing to own makers

Fluids and rules

PFAS limits and costly two-phase fluids

A moving target

Racks heading from 120 kW to 600 kW to 1 MW

ARC RESEARCH VIEW — A VIEW, NOT ADVICE

What it means for AI GPUs

Single-phase is the standard now

NVIDIA and OCP specs point the same way

Two-phase is the headroom bet

Matters more as racks near 600 kW to 1 MW

Fluids gate two-phase

Watch the EU opinion due by end-2026

Cooling sets sustained speed

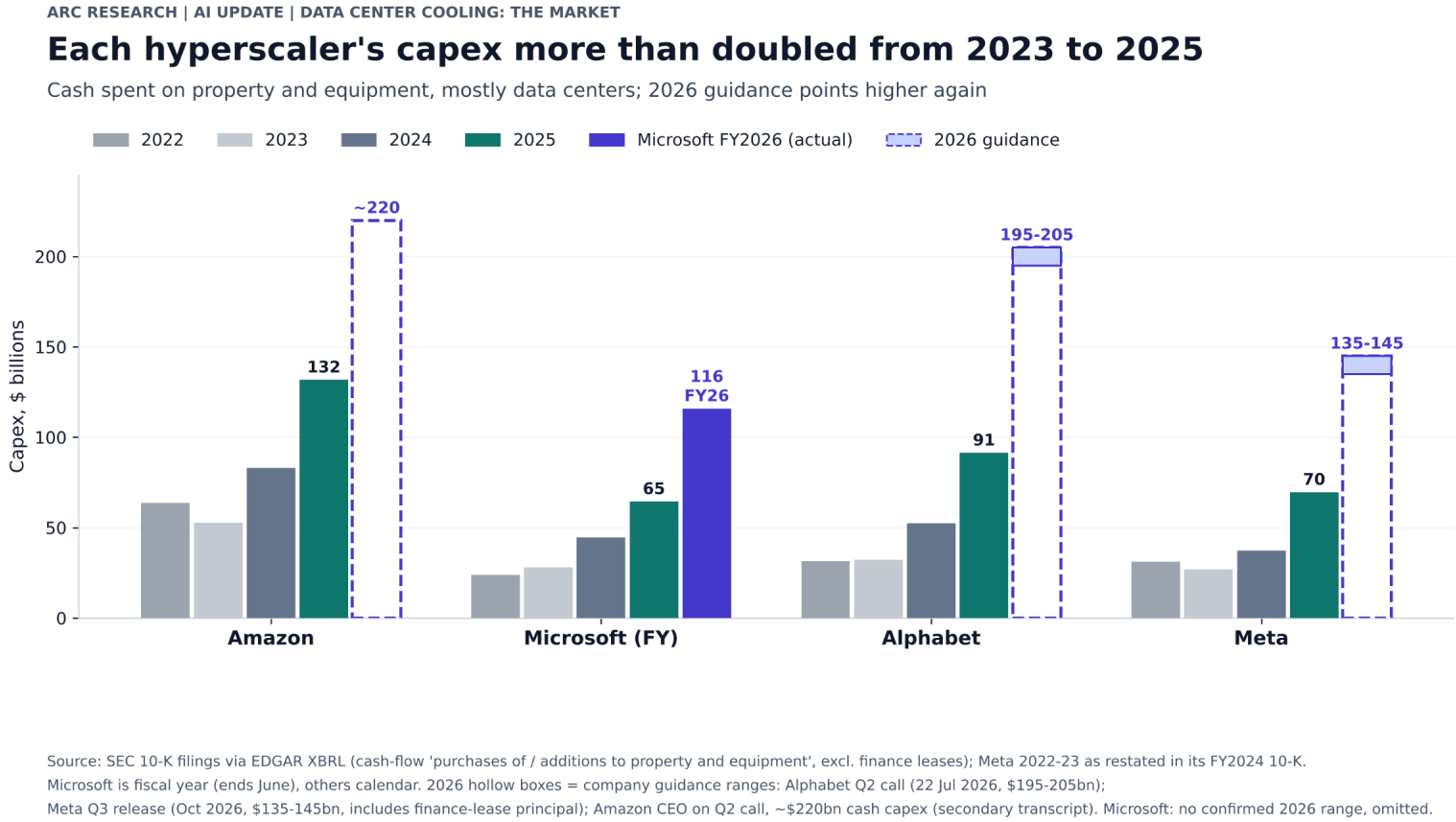
Cooler GPUs hold clocks; denser racks follow

06

The market angle

Up next: Who’s spending · How big it gets · Who’s winning · Who’s buying · ARC view

Who pays: Big Tech’s capex boom



Four companies spent ~$357bn in 2025

2026 guides point higher again

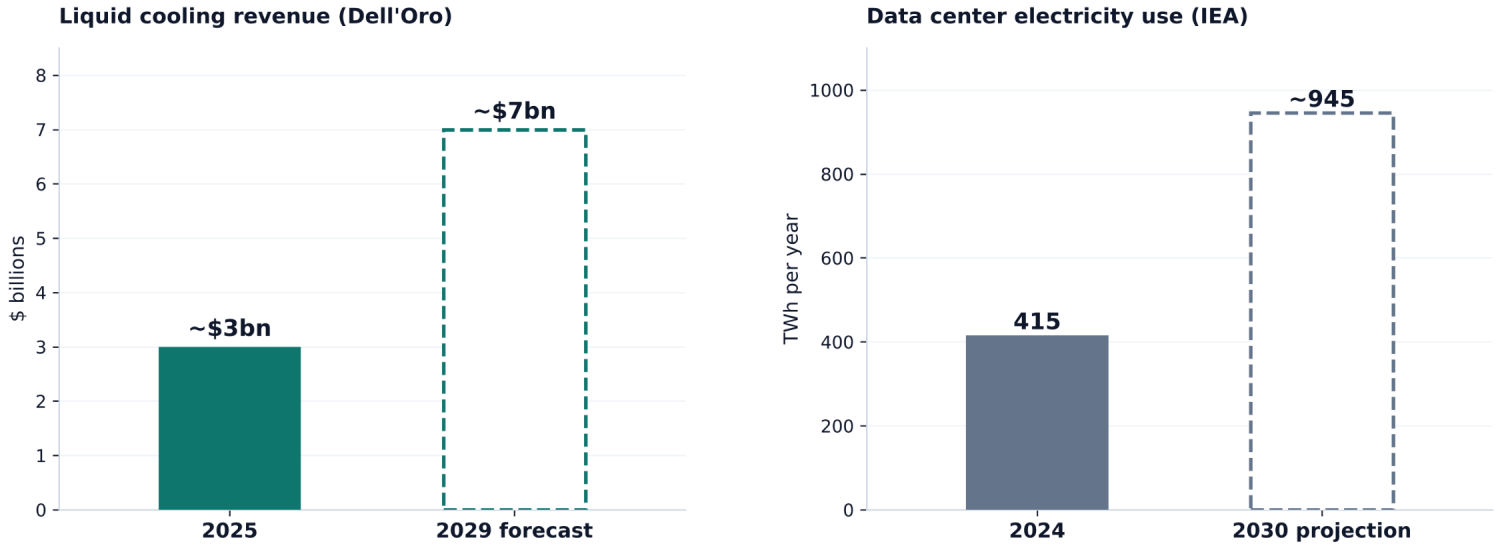
Every new AI rack needs liquid cooling

How big it gets: ~$3bn to ~$7bn

ARC RESEARCH | AI UPDATE | DATA CENTER COOLING: THE MARKET

Liquid cooling is a ~$3bn market heading for ~$7bn

Dell'Oro expects liquid cooling revenue to more than double by 2029 as data center power use doubles



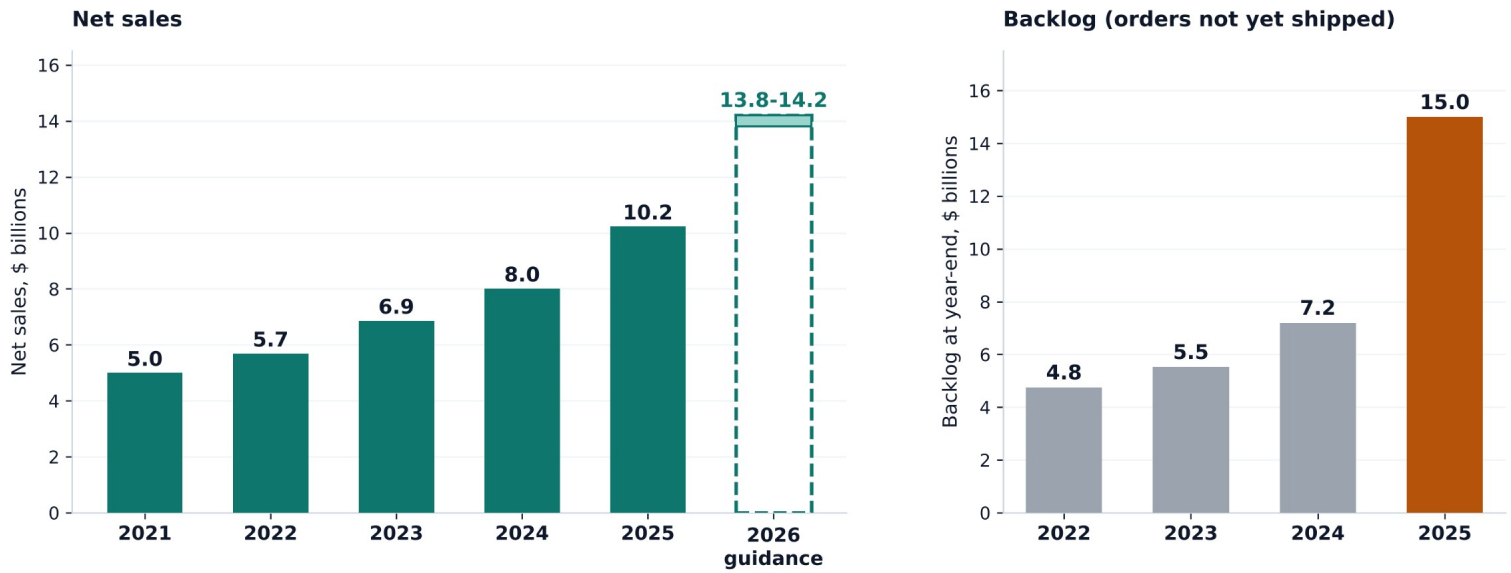
Source: Dell'Oro Group, 8 Jan 2026 (liquid cooling manufacturer revenue: 'close to $3B' in 2025, ~$7B by 2029; an earlier Dell'Oro release, 10 Dec 2025, put 2025 DLC at >$2B). IEA, Energy and AI (2025): data center electricity 415 TWh in 2024, ~945 TWh projected for 2030. Hollow dashed = forecast or projection.

Liquid cooling to more than double by 2029

Data center power use to double by 2030

Every saved watt of cooling is worth more

Who’s winning: Vertiv’s numbers



Sales doubled, 2021 to 2025

Backlog doubled in 2025 alone

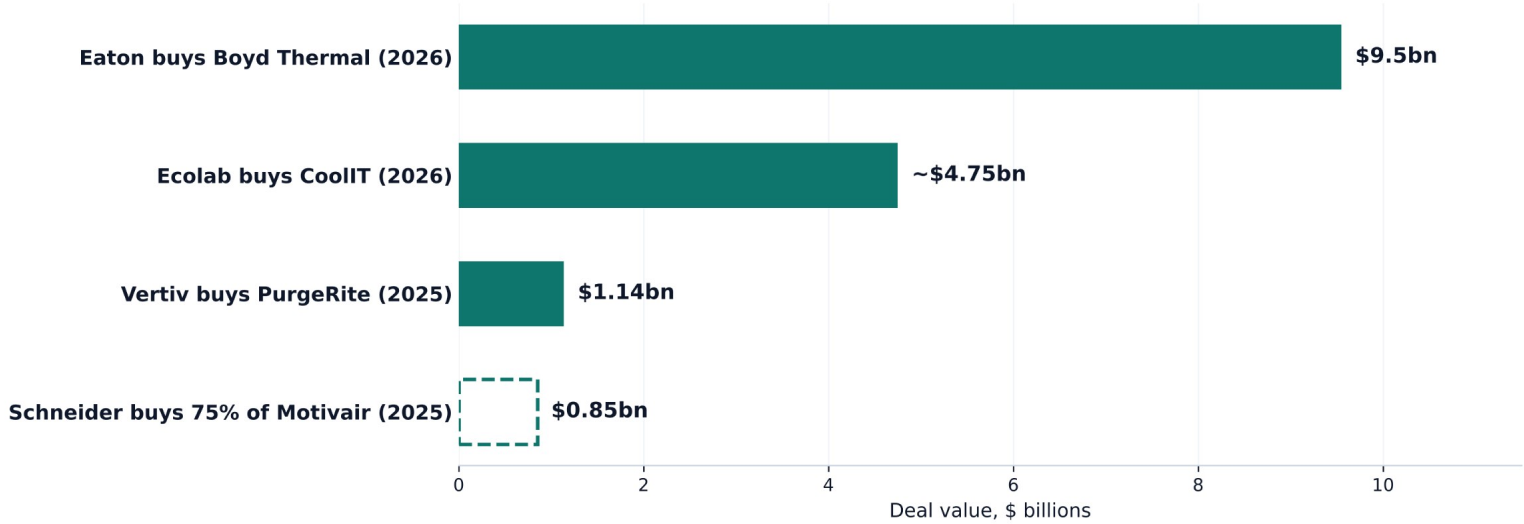
Q3 results due 28 Oct

Who’s buying: cooling M&A

ARC RESEARCH | AI UPDATE | DATA CENTER COOLING: THE MARKET

Industrial giants are paying up to own liquid cooling

Eaton paid 22.5x Boyd Thermal's 2026 EBITDA; Ecolab paid 29x CoolIT's next-12-month EBITDA



Source: Eaton release (3 Nov 2025) and Q2 2026 10-Q (Boyd Thermal, closed 12 Mar 2026, $9.5bn); Ecolab release (19 Mar 2026) and Q2 2026 10-Q (CoolIT, ~$4.75bn, closed 2 Jul 2026); Vertiv Q2 2026 10-Q (PurgeRite, $1.14bn net, closed Dec 2025); DCD, Oct 2024 (Schneider: $850m for 75% of Motivair; Schneider confirms the 75% stake but its release gives no price).
Solid = price in company filing or release; hollow dashed = price press-reported. Multiples are as stated by the acquirers.
Deals with undisclosed prices (Trane-LiquidStack, Daikin-Chilldyne, Jabil-Mikros) not shown.

~$16bn across four deals

Priced at 22–29× EBITDA

Buyers pay for future growth

ARC RESEARCH VIEW — A VIEW, NOT ADVICE

What achieving liquid cooling is worth

Owners: more compute per megawatt

Less power to cooling, denser racks, faster GPUs

Suppliers: a niche growing fast

~$3bn to ~$7bn by 2029; Vertiv backlog $15bn

Prices already assume growth

Deals at 22–29× EBITDA leave little room for misses

Risks to watch

Capex cycle, PFAS rules, Q3 results this week

Key terms

Term	In plain words
Single-phase	Coolant stays liquid and simply warms up
Two-phase	Coolant boils on the hot part, then condenses back
Cold plate (DLC)	Metal block on the chip with coolant flowing inside
Immersion	Whole server dunked in a non-conductive fluid
CDU	Pump and heat exchanger between rack and building
PUE	Total site power ÷ computer power (1.0 = no overhead)
PFAS	“Forever chemicals”: long-lasting fluorinated compounds
Throttling	A hot GPU slowing itself down to stay safe

Sources

Vendor and filed

NVIDIA V100 datasheet

NVIDIA A100 · H100 · H200

NVIDIA: Inside Blackwell Ultra

NVIDIA: Blackwell liquid cooling

NVIDIA: Inside Rubin platform

NVIDIA: 800 V HVDC

NVIDIA NVSentinel runbook

AMD MI300X · MI355X

Google Cloud: 1 MW racks

3M: exit PFAS by 2025

Chemours Opteon 2P50

Industry bodies and research

OCP: About · Cooling Environments

OCP: ACS sub-project (2018)

OCP Immersion Reqs Rev 1.0

OCP APAC Summit (Aug 2025)

OCP Summit 2026 in numbers

Uptime Global Survey 2026

Uptime Cooling Survey 2024

IEA: Energy and AI

ORNL: 2P50 pool boiling

Press-reported and vendor blogs

DCD: Rubin Ultra 600 kW

DCD: Chemours 2P50 launch

Network World: OCP 2026

White & Case: EU PFAS

Schneider Electric blog

GRC: two- vs single-phase

Sources: markets and players

Filed and company-published

Amazon capex (SEC XBRL) · Q2 2026 release

Microsoft capex (SEC XBRL)

Alphabet capex (SEC XBRL) · Q2 2026 call

Meta capex (SEC XBRL) · Q3 2026 release

Vertiv FY2025 10-K · Q2 2026 release

Eaton: Boyd Thermal deal · Q2 2026 10-Q

Ecolab: CoolIT deal · Q2 2026 10-Q

Modine Q4 FY2026 results

Research firms and press-reported

Dell'Oro: liquid cooling to ~$7bn by 2029

Dell'Oro: DCPI $10.9bn in 4Q25

Dell'Oro: data center capex 2025–26

IEA: Energy and AI

DCD: Schneider–Motivair

Facilities Dive: Schneider Q3 2026

Amazon Q2 2026 call transcript

Players and challenges

DCF: Microsoft microfluidics · DCD: Accelsius

Capacity: Submer · Fierce: Trane–LiquidStack

Uptime: DLC challenges · Network World: OCP 2026